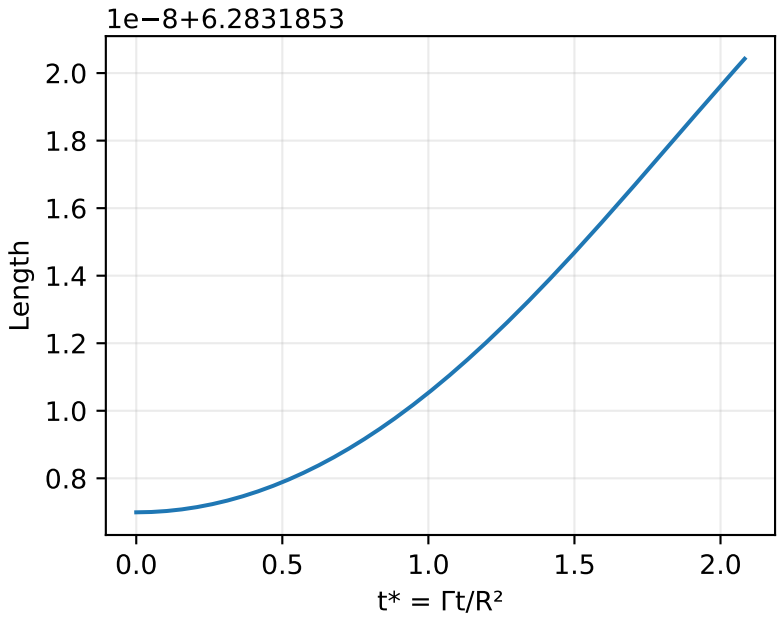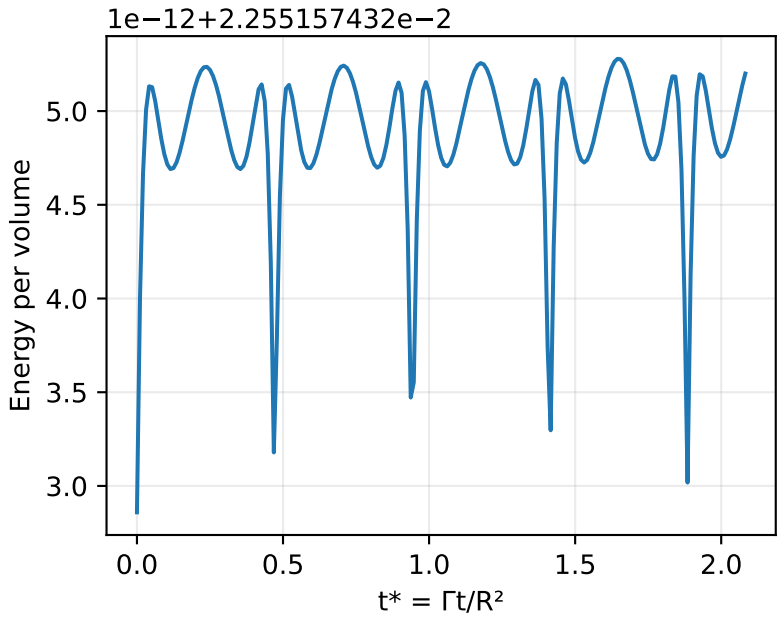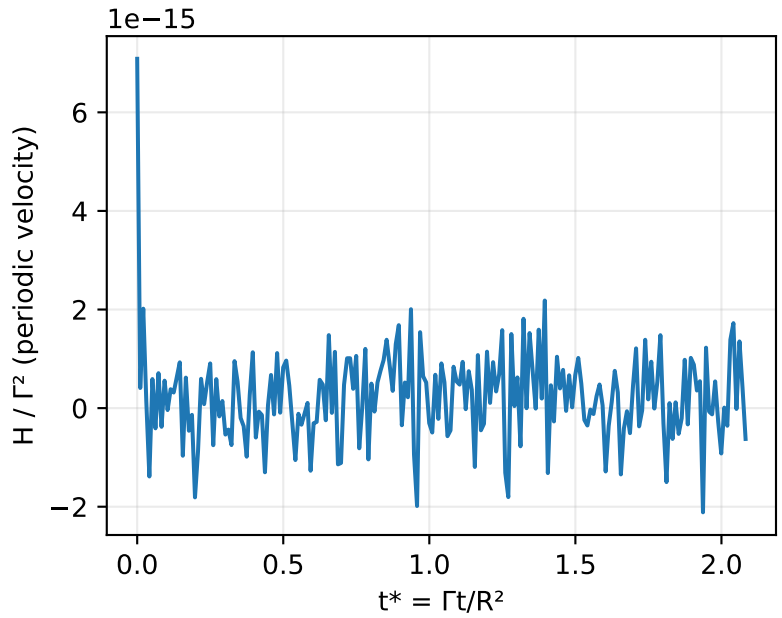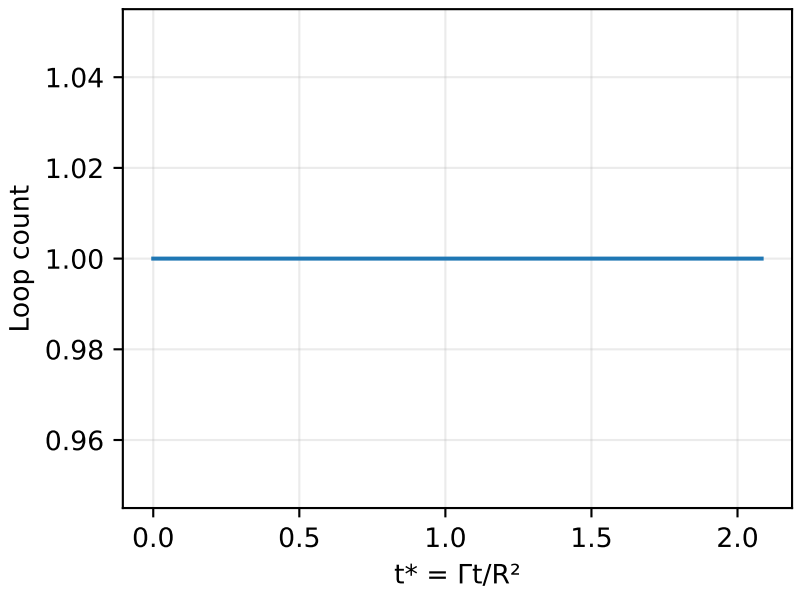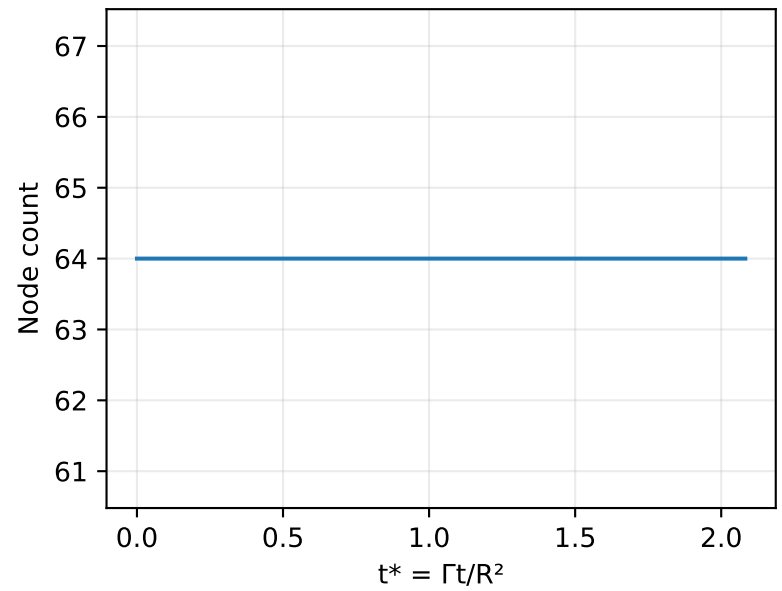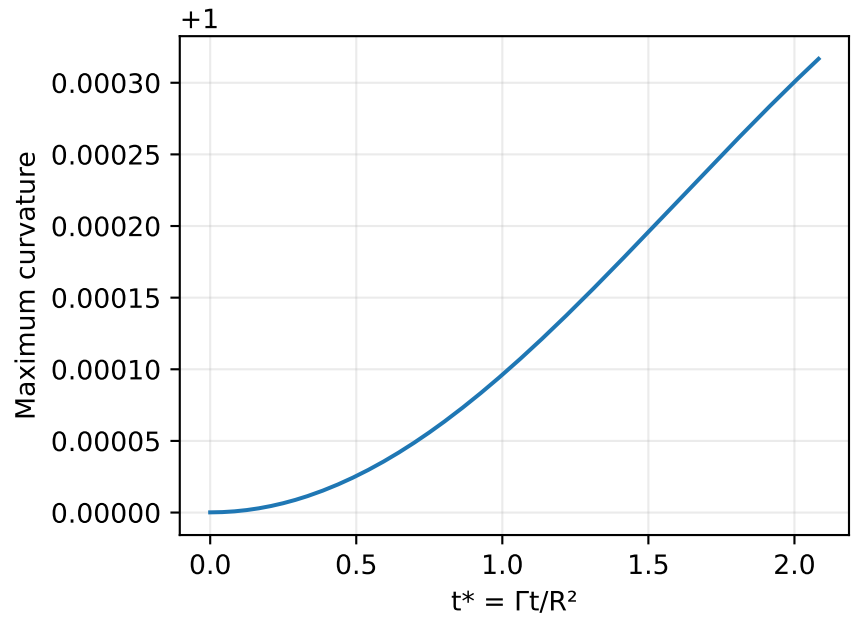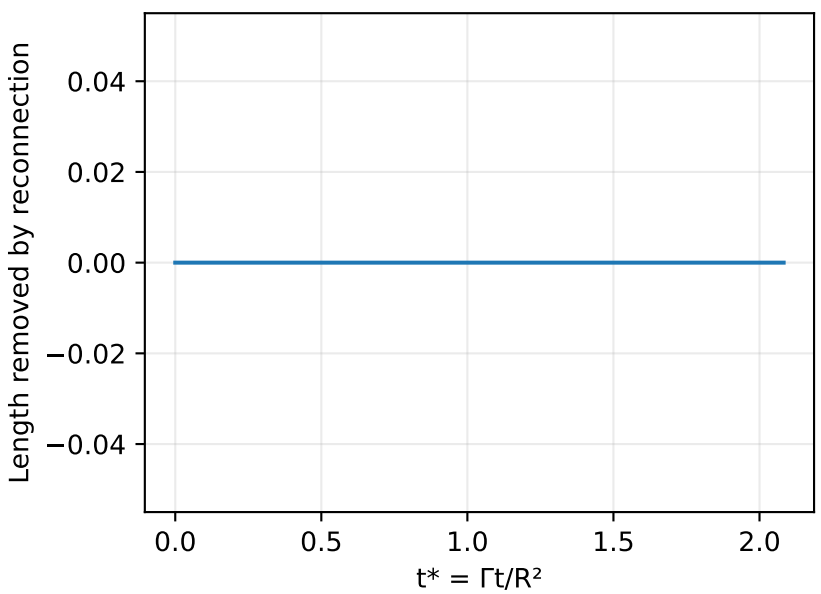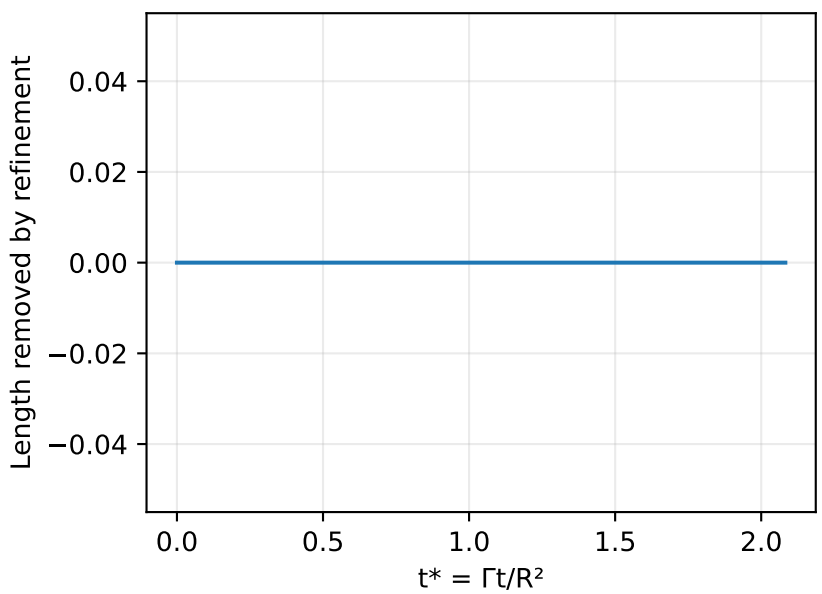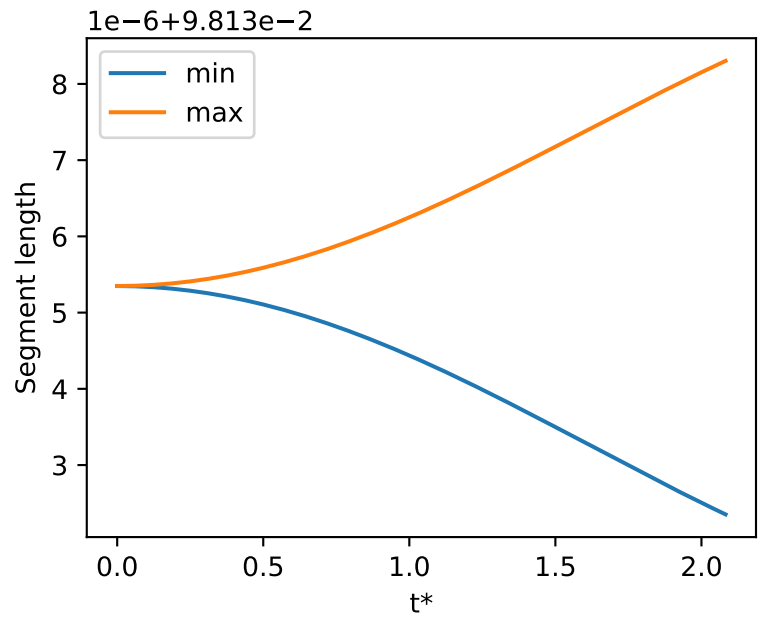ring_wide · a=0.1171875 · no mutual friction